The Failure Point of Systems of Linearized Differential Equations

How far from equilibrium does the linearization of a two-variable energy-balance climate model remain a valid predictor of the system's behaviour?

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1 Introduction

Substituting back into $\alpha(T)$ gives:

$$\alpha(T) = \frac{1}{2}\left[(\alpha_W + \alpha_C) + (\alpha_W - \alpha_C)\left(\frac{u}{\omega} - \frac{u^3}{3\omega^3} + O\left(\frac{u^5}{\omega^5}\right)\right)\right].$$

Then, substituting this into the energy model and substituting $T = T_R + u$:

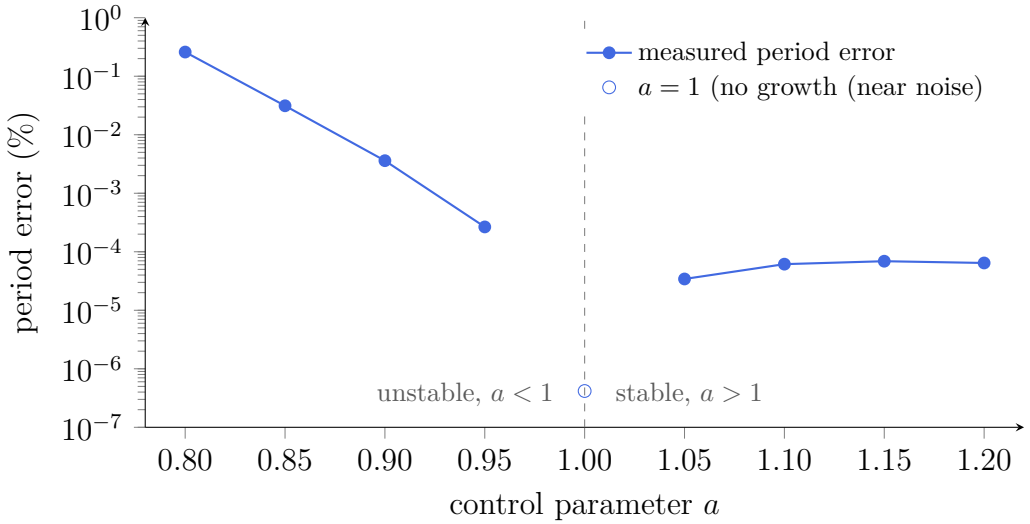
$$C\frac{dT}{dt} = Q\left[1 - \frac{1}{2}\left[(\alpha_W + \alpha_C) + (\alpha_W - \alpha_C)\left(\frac{u}{\omega} - \frac{u^3}{3\omega^3} + O\left(\frac{u^5}{\omega^5}\right)\right)\right]\right] - (A + B\,(T_R + u)).$$

Then distributing Q:

$$C\frac{dT}{dt} = \left[Q + \left[-\frac{Q}{2}(\alpha_W + \alpha_C) - \frac{Q}{2}(\alpha_W - \alpha_C)\left(\frac{u}{\omega} - \frac{u^3}{3\omega^3} + O\left(\frac{u^5}{\omega^5}\right)\right)\right]\right] - (A + B\,(T_R + u)).$$

$$C\frac{dT}{dt} = Q - \frac{Q}{2}(\alpha_W + \alpha_C) - \frac{Q\,(\alpha_W - \alpha_C)}{2\omega}u + \frac{Q\,(\alpha_W - \alpha_C)}{6\omega^3}u^3 + O\left(\frac{u^5}{\omega^5}\right) - A - B\,(T_R + u)$$

$$C\frac{du}{dt} = \left[Q - \frac{Q}{2}(\alpha_W + \alpha_C) - A - BT_R\right] + \left[-\frac{Q\,(\alpha_W - \alpha_C)}{2\omega} - B\right]u + \frac{Q\,(\alpha_W - \alpha_C)}{6\omega^3}u^3 + O\left(\frac{u^5}{\omega^5}\right)$$



$$\tanh x = x - \frac{x^3}{3} + O(x^5) \quad \Rightarrow \quad \tanh \frac{u}{\omega} = \frac{u}{\omega} - \frac{u^3}{3\omega^3} + O\left(\frac{u^5}{\omega^5}\right)$$

2 Methodology

3 Evaluation

3.1 Limitations

3.2 Strengths

4 Conclusion